

Theoretical Bounds and Mechanisms for Neural Scaling Laws

Research synthesis

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Abstract

Neural scaling laws are empirical regularities: over substantial but finite ranges, held-out loss often decreases approximately as a power of model size, training data, or compute. This survey organizes exact results in specified models, controlled-model explanations, and mechanistic transfers to modern neural networks. Rigorous examples include Gaussian rate-distortion asymptotics, minimax nonparametric rates, kernel and linear-regression learning curves under spectral assumptions, associative-memory models, and approximation lower bounds. Manifold tiling, discrete “quanta,” and feature superposition provide additional mechanisms with measurable assumptions. A 2026 theory of Cagnetta, Raventós, Ganguli, and Wyart connects data-limited language-model exponents to measured conditional-entropy and token-correlation decay under an explicit horizon-limited condition, offering a parameter-free prediction of real-corpus exponents. These frameworks are complementary: coding rate, geometric resolution, spectral resolution, statistical horizon, and discrete memory are different resources, and identifying which one governs a trained model is an empirical question. Throughout, we state the scope of each result, derive its power law when possible, and identify the measurements that can make it predictive.

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1 Scope, terminology, and standards of evidence

Let N denote a model-size measure, D the number of training examples or tokens, and C a training-compute estimate. A common empirical fit is

$$L(N, D) = E + AN^{-\alpha_N} + BD^{-\alpha_D}. \quad (1)$$

Equation (1) is an *ansatz*, not a general law. Its fitted constants depend on the data distribution, tokenizer, context length, model family, optimization protocol, compute accounting, and fitting window. Hestness et al. [16] and Rosenfeld et al. [34] documented predictable power-law error curves across several domains before the language-model literature consolidated around Equation (1); Kaplan et al. [23] and Hoffmann et al. [18] established its practical value for compute allocation, and later replication and ablation work showed that fitting and training choices matter [4, 31].

Notation. Symbols are reused across the literature with incompatible meanings, so this survey fixes the following conventions and flags departures explicitly. Throughout, N is model size, D is data size, C is compute, and n is a generic sample size or block length inside a stylized model. Exponents attached to a resource carry a subscript (α_N, α_D); bare Greek letters are local to the subsection that introduces them. Where a cited paper uses P for tokens, or M and N for parameters and samples respectively, the statement is translated into the conventions above and the original notation is noted.

This survey uses the following evidence hierarchy.

1. **Rigorous conditional result:** a theorem for a fully specified source, function class, loss, algorithm, or asymptotic regime.
2. **Controlled-model explanation:** an analytic or numerical result in a teacher–student, kernel, random-feature, or discrete toy model.
3. **Mechanistic explanation:** a proposed transfer from a controlled model to trained finite neural networks, accompanied by testable assumptions.
4. **Empirical fit:** a description of observed runs over a stated range, with no claim of asymptotic universality.

How to read the frameworks

Rate–distortion, data-manifold geometry, neural-tangent-kernel spectra [20], and Zipf-like distributions [47] describe different routes to scaling. Each framework becomes predictive for modern language models when its assumptions are connected to training and its governing exponents are measured independently of the loss curve.

2 What the empirical laws establish

Hoffmann et al. fit Equation (1)—writing α and β for the exponents this survey calls α_N and α_D —to their language-model experiments, reporting the rounded values $E = 1.69$, $A = 406.4$, $B = 410.7$, $\alpha_N = 0.34$, and $\alpha_D = 0.28$ [18]. Minimizing the fitted expression at fixed $C \simeq 6ND$ gives the stationarity condition $\alpha_N AN^{-\alpha_N} = \alpha_D BD^{-\alpha_D}$, hence

$$N_{\text{opt}} \propto C^{\alpha_D/(\alpha_N+\alpha_D)}, \quad D_{\text{opt}} \propto C^{\alpha_N/(\alpha_N+\alpha_D)}, \quad (2)$$

which for the rounded exponents above is $C^{0.45}$ and $C^{0.55}$, against the $C^{0.46}$ and $C^{0.54}$ that Hoffmann et al. report from their unrounded fit: nearly equal power-law growth of compute-optimal

model size and token count. This is the source of the Chinchilla prescription, and Equation (2) makes its dependence on the fitted exponent ratio α_N/α_D transparent. The estimate is specific to a data mixture, model family, and training recipe. Besiroglu et al. reconstructed the underlying data and found appreciable uncertainty in the third estimation procedure and tension with the first two procedures [4]. Porian et al. then showed how last-layer compute accounting, warmup duration, and scale-dependent optimizer tuning reconcile much of the Kaplan–Chinchilla difference [31]. Together, these studies provide a more informative uncertainty and sensitivity analysis for compute-optimal allocation.

Two distinctions are essential.

- **Loss versus capability metrics.** Smooth average log loss can coexist with sharp changes in thresholded accuracy. Reported “emergent” capabilities [44] may reflect both changes in the predictive distribution and the geometry of the evaluation metric: Schaeffer et al. show that discontinuous and nonlinear metrics such as exact-match accuracy can turn smooth per-token improvements into apparent jumps, while continuous metrics on the same runs often scale smoothly [35]. Stating the measurement model therefore makes capability-scaling claims more precise.
- **Interpolation versus extrapolation.** A straight line in a log–log plot supports a power law over the observed range. Broken-power-law fits capture the possibility that curves change slope, plateau, or steepen outside that window [7], and provide a useful extension when more than one regime is visible.

3 Information-theoretic foundations and connections

3.1 Cross-entropy and the Bayes floor

For a fixed evaluation distribution P^* , context variable Z_t (the conditioning information available at step t), next token X_{t+1} , and predictive distribution Q , expected log loss decomposes exactly as

$$\begin{aligned} \mathbb{E}_{P^*}[-\log Q(X_{t+1} | Z_t)] &= H_{P^*}(X_{t+1} | Z_t) \\ &\quad + \mathbb{E}_{Z_t} D_{\text{KL}}(P^*(\cdot | Z_t) \| Q(\cdot | Z_t)). \end{aligned} \quad (3)$$

Because the divergence term is nonnegative and vanishes only at $Q = P^*$, the conditional entropy is exactly the Bayes risk for log loss with that conditioning information. For a stationary source on a finite alphabet, $H(X_{t+1} | X_1^t)$ is nonincreasing in t and therefore converges to the entropy rate $h = \lim_t H(X_{t+1} | X_1^t)$ [11, 36]. Real corpora are not stationary, and h is then a property of an assumed idealization rather than a measurable constant.

The fitted constant E in Equation (1) need not equal h , and each of the following gaps is one-directional. A finite context window of length T has floor $H(X_{t+1} | X_{t-T+1}^t) \geq h$; restricted architecture, optimization error, regularization, or distribution shift can raise the asymptote further; and fitting E jointly with A, B, α_N, α_D over a finite window estimates the intercept of an extrapolated curve, not a limit that was observed. Moreover, entropy per token depends on the tokenizer, so E is not even dimensionally comparable across tokenizations without rescaling by tokens per character. Shannon’s character-prediction experiment gave bounds and estimates on a character scale [37]; modern token losses cannot be compared to those numbers without converting both measurements on the same corpus. Compression and extrapolation estimates also depend on corpus and modeling assumptions [42].

3.2 Channel reliability and predictive loss

For a discrete memoryless channel used below capacity, coding theorems yield exponential-in-blocklength behavior for achievable or optimal decoding error in appropriate rate regimes. This

concerns communication over repeated noisy channel uses, whereas neural scaling curves usually concern expected prediction loss as a function of resources. A useful connection therefore needs an explicit correspondence between channel uses, model resources, and the chosen loss.

Rate–distortion theory provides a closer mathematical analogy because it studies the smallest expected distortion attainable under a representation-rate constraint. The following example is exact.

Theorem 3.1 (Gaussian reverse water-filling example). Let X be a zero-mean Gaussian random element of a separable Hilbert space whose covariance eigenvalues are

$$\lambda_k = ck^{-(1+\rho)}, \quad c > 0, \rho > 0, \quad (4)$$

so that $\sum_k \lambda_k < \infty$. Under squared-error distortion the distortion–rate function is given parametrically by reverse water-filling,

$$D(\theta) = \sum_{k \geq 1} \min(\theta, \lambda_k), \quad R(\theta) = \frac{1}{2} \sum_{k \geq 1} \log_2^+ \frac{\lambda_k}{\theta}, \quad (5)$$

with R measured in bits. As $R \rightarrow \infty$, $D(R) \asymp R^{-\rho}$.

Proof. Only the modes with $\lambda_k > \theta$ are active, so their number is $K = K(\theta) = \lfloor (c/\theta)^{1/(1+\rho)} \rfloor \asymp (c/\theta)^{1/(1+\rho)}$; equivalently $\theta \asymp cK^{-(1+\rho)}$ and $\lambda_k/\theta \asymp (K/k)^{1+\rho}$ for $k \leq K$. Hence

$$R(\theta) \asymp \frac{1+\rho}{2} \sum_{k=1}^K \log_2 \frac{K}{k} = \frac{1+\rho}{2} \log_2 \frac{K^K}{K!} = \frac{(1+\rho) \log_2 e}{2} K + O(\log K) \quad (6)$$

by Stirling’s formula, so $R(\theta) \asymp K(\theta)$: each additional active mode buys a bounded number of bits. For the distortion, the active modes contribute $K\theta \asymp cK^{-\rho}$ and the inactive tail contributes $\sum_{k>K} ck^{-(1+\rho)} \asymp (c/\rho)K^{-\rho}$, so

$$D(\theta) = K(\theta)\theta + \sum_{k>K(\theta)} ck^{-(1+\rho)} \asymp K(\theta)^{-\rho} \asymp \theta^{\rho/(1+\rho)}. \quad (7)$$

Eliminating θ through $R \asymp K$ yields $D(R) \asymp R^{-\rho}$. $\square$

Two features of Theorem 3.1 are worth isolating, because they recur in every mechanism surveyed below. First, the power law comes entirely from the polynomial tail of the spectrum: a spectrum with a geometric tail would give exponential decay instead, as the finite-dimensional case below shows. Second, the exponent is the spectral tail exponent minus one, a property of the source alone—not of the coder, the loss scale, or the dimension of any ambient space.

For a fixed d -dimensional nonsingular Gaussian source, all d modes are active at sufficiently high rate and $D(R)$ is proportional to $2^{-2R/d}$ up to a covariance-dependent constant [3, 11].

Remark 3.1 (Interpretation for neural scaling). The example shows that a heavy spectral tail produces polynomial distortion–rate behavior. Connecting neural parameter count N with communication rate R requires an encoder/decoder, a bit-precision or description-length budget, a distortion matching the training loss, and an argument that training approaches the corresponding optimum. Those additions would turn the rate–distortion analogy into a predictive neural model and would distinguish it from the other power-law mechanisms discussed below.

3.3 Metric entropy and minimax estimation

For a function class $\mathcal{P}$ equipped with a metric ϱ , write

$$\mathcal{H}(\varepsilon, \mathcal{P}, \varrho) = \log \mathcal{N}(\varepsilon, \mathcal{P}, \varrho) \quad (8)$$

for its covering entropy, where $\mathcal{N}$ is the ε -covering number. Yang and Barron relate minimax rates to global and local metric entropy through information-theoretic upper bounds and testing lower bounds [45]; see also [43] for the accompanying reduction and two-point techniques. The mnemonic balance

$$\mathcal{H}(\varepsilon_n) \asymp n\varepsilon_n^2 \quad (9)$$

is valid in many regular nonparametric problems when the loss, KL control, and local packing/testing conditions link global complexity to local distinguishability. Keeping these conditions explicit is important for classes whose local and global entropies have different orders.

For standard Sobolev classes of smoothness s on a d -dimensional compact domain, $\mathcal{H}(\varepsilon) \asymp \varepsilon^{-d/s}$ in the relevant norm, and Equation (9) then yields the familiar squared-risk rate

$$\varepsilon_n^2 \asymp n^{-2s/(2s+d)}. \quad (10)$$

Here n is the number of i.i.d. observations, which corresponds to D rather than N in the notation of Equation (1). This is a minimax result for a specified function class and observation model. It becomes a language-model prediction after identifying the effective class, metric, dependence structure, and estimator; the resulting exponent depends on the ratio s/d , which suggests a concrete program for operationalizing smoothness and dimension in next-token prediction.

3.4 Information-theoretic model/data allocation

Jeon and Van Roy develop a Bayesian information-theoretic decomposition for prediction with a compressed or misspecified latent model [21]. In schematic form, their bound splits the average excess log loss over T observations into

$$\mathcal{E}_T \lesssim \underbrace{\frac{I(F; \tilde{F})}{T}}_{\text{estimation/information term}} + \underbrace{\mathbb{E} D_{\text{KL}}(P_F(\cdot | X) \| P_{\tilde{F}}(\cdot | X))}_{\text{misspecification term}}. \quad (11)$$

Here F is the data-generating function, $\tilde{F}$ its compressed or misspecified surrogate, T the number of observations, and $\mathcal{E}_T$ the per-observation excess loss; the first term is the cost of identifying $\tilde{F}$ from data and the second the cost of $\tilde{F}$ not being F , so the two move in opposite directions as the surrogate is made richer. For their infinite-width two-layer-network data-generating process, optimizing a derived upper bound under a product compute proxy yields an optimal relation between data size and model size that is linear up to logarithmic factors. This rigorous Bayesian result is consistent with the empirical Chinchilla ratio. Extending it to SGD-trained transformers calls for an optimization model and a bridge from transformer architecture to the paper’s latent-function compression; the model-specific analysis, rather than the generic mutual-information rate bounds alone, supplies the compute-allocation conclusion.

4 Language statistics and data-limited scaling

4.1 Hilbert-type entropy growth

A precise version of a Hilbert-type ansatz writes block entropy as

$$H(X_1^n) = hn + cn^\delta + o(n^\delta), \quad 0 < \delta < 1. \quad (12)$$

Taking first differences gives

$$H(X_{n+1} | X_1^n) - h \asymp n^{-(1-\delta)} \quad (13)$$

when the asymptotic expansion is sufficiently regular—that is, when the expansion may be differenced term by term, which is an assumption about the source and not a consequence

of Equation (12) alone. Adjacent-block mutual information, which for a stationary source equals $I(X_1^n; X_{n+1}^{2n}) = 2H(X_1^n) - H(X_1^{2n})$, also inherits an n^δ contribution under Equation (12). These relationships follow from the block-entropy ansatz; they should not be inferred merely by differentiating an empirical mutual-information plot [17, 42]. Hilberg’s original claim was a reinterpretation of Shannon’s experimental data over a very short range of n ; Dębowski’s relaxed form, which replaces the equality by upper and lower power-law bounds, is the version with meaningful empirical support and the one connected to theorems about long-memory sources [13].

4.2 Cagnetta–Raventós–Ganguli–Wyart (2026)

Cagnetta et al. study a data-limited regime in which model size and optimization are intended not to be bottlenecks [9]; they write P for the token budget, which is D here, and $C(n)$ for the covariance matrix written $\Sigma(n)$ below. Their prediction uses two measured dataset statistics, both indexed by a context length or lag n :

$$H_n - H_\infty \asymp n^{-\gamma}, \quad \|\Sigma(n)\|_{\text{op}} \asymp n^{-\beta}, \quad (14)$$

where $H_n = H(X_{t+1} | X_{t-n+1}^t)$ is the next-token conditional entropy at context length n and $\Sigma(n)$ is the token–token covariance matrix at lag n . The argument has two steps. First, since the empirical covariance at lag n is estimated with noise $O(D^{-1/2})$, a correlation of size $n^{-\beta}$ is statistically resolvable only while $n^{-\beta} \gtrsim D^{-1/2}$, so the largest usable horizon is

$$n^*(D) \asymp D^{1/(2\beta)}. \quad (15)$$

Second, if within-horizon excess errors decay faster than the error caused by truncating the context at n^* , the loss is set by the conditional-entropy gap at that horizon:

$$\mathcal{L}(D) - H_\infty \asymp (n^*(D))^{-\gamma} \asymp D^{-\gamma/(2\beta)}, \quad \alpha_D = \frac{\gamma}{2\beta}. \quad (16)$$

The fast within-horizon learning condition is substantive rather than technical: it delineates models expressive enough to exploit the horizon they can resolve from shallow networks, kernels, or n -gram methods for which within-horizon estimation may remain the bottleneck. Their reported estimates are

$$\begin{aligned} \text{TinyStories: } \gamma &= 0.325 \pm 0.003, \quad \beta = 0.88 \pm 0.06 \quad \Rightarrow \quad \alpha_D = 0.185 \pm 0.013, \\ \text{WikiText: } \gamma &= 0.265 \pm 0.016, \quad \beta = 0.94 \pm 0.16 \quad \Rightarrow \quad \alpha_D = 0.141 \pm 0.025, \end{aligned} \quad (17)$$

with the intervals obtained by bootstrapping and standard error propagation, and with no parameters fitted to the learning curves themselves. Empirical exponents from GPT-2- and LLaMA-style transformers fall inside those intervals over the tested ranges, and the loss curves collapse under the rescaling $D \mapsto D/n^{2\beta}$, $\mathcal{L}_n \mapsto n^\gamma \mathcal{L}_n$ implied by the derivation. Their γ estimates are somewhat above the value near 0.23 obtained by compression-based extrapolation [42], on different corpora.

This result is unusually direct because it predicts an exponent from independently measured corpus statistics rather than from a synthetic data model, and because both inputs are falsifiable on any new corpus. Equation (16) also has discriminating content: corpora sharing γ but differing in correlation decay β should have different data exponents. Three points guide interpretation. The power laws are fitted over finite ranges, and the WikiText correlation exponent carries a ± 0.16 uncertainty that dominates its prediction interval. The estimates of H_n use trained models as upper bounds, so future entropy estimators can further calibrate γ . Finally, the LLaMA–TinyStories collapse is visually sharpest near $\beta \simeq 0.65$, outside the independently measured range; the authors identify this as a useful finite-range or architecture-dependent effect to investigate. The theory directly addresses data-limited scaling, leaving parameter and compute scaling as natural extensions.

4.3 Controlled tests of data structure

Barkeshli, Alfarano, and Gromov train small transformers on random walks over graphs with tunable structure and on progressively simplified generative models of language [2]. Their experiments recover neural scaling curves even for Erdős–Rényi graph data without a power-law correlation spectrum, while language-model exponents evolve monotonically as the generator is simplified. This supplies a valuable complementary result: heavy-tailed data statistics can determine exponents in models such as Equation (16), but neural training can also produce finite-range scaling through other mechanisms. Controlled generators offer a promising intervention-based method for identifying which data statistic sets an exponent.

5 Geometry, regimes, and approximation limits

5.1 The manifold-tiling mechanism

Suppose a smooth target on a compact d -dimensional manifold is approximated by affine pieces on K cells of typical diameter $K^{-1/d}$. If the pointwise Taylor remainder is second order and loss is squared error, then

$$|f - \hat{f}| = O(K^{-2/d}), \quad \mathbb{E}|f - \hat{f}|^2 = O(K^{-4/d}). \quad (18)$$

If a network provides $K \asymp N$ effective cells, Equation (18) suggests $\alpha_N \approx 4/d$ [38].

The last step— $K \asymp N$ —is the modeling bridge from geometry to parameters. Deep ReLU networks can have exponentially more linear regions than parameters [28]; learned regions may be nonuniform, task-aligned, and scale dependent; and classification cross-entropy differs from global squared regression. Measuring the number, occupancy, and curvature of effective cells would therefore make the mechanism directly testable.

Sharma and Kaplan present the transfer to realistic networks as a mechanistic proposal, and their language-model evidence supports the inequality $d \geq 4/\alpha_N$. For GPT-type models with $\alpha_N \approx 0.076$ one has $4/\alpha_N \approx 53$, while intrinsic-dimension estimates from last-token activations exceed 90. Activations drawn from one contiguous 1024-token passage instead have estimated dimension about 7, an order of magnitude below the estimate across passages. Together these measurements suggest a scale-dependent hierarchy of manifolds rather than a single global d , which may be more informative for future theories. The frequently repeated range $d \approx 10$ –30 is not a result reported in this paper.

5.2 Four scaling regimes and their crossover

Bahri et al. distinguish variance- and resolution-limited behavior along both model and data axes, producing four regimes [1]:

Varied resource	Limiting mechanism	Behavior in their analysis
data D	finite-sample variance near an infinite-data limit	correction $O(D^{-1})$
width w	finite-width variance near an infinite-width limit	correction $O(w^{-1})$
data D	resolution of a smooth d -manifold	exponent proportional to $1/d$
parameters or features P	resolution of a smooth d -manifold	exponent proportional to $1/d$

The D^{-1} and width results follow from the existence of a well-behaved infinite-data or infinite-width limit plus averaging, and are observed with exponent very close to 1 in the controlled experiments. The four regimes describe alternative bottlenecks rather than additive contributions. In the resolution-limited regime, Theorems 2 and 3 assume that the target, model, and loss are

Lipschitz and that the model interpolates the training points; they yield $L = O(D^{-1/d})$ and $L = O(P^{-1/d})$, where P is a parameter or feature count, through a nearest-neighbour distance bound on a d -manifold. The paper complements these rigorous bounds with the typical-case estimate $D^{-c/d}$, $c \geq 2$, under analyticity of the target and loss, with c determined by the leading nonvanishing Taylor order. Smooth harmonic crossover fits [7] provide a practical empirical bridge between the regimes.

5.3 Scope of approximation and learning bounds

Continuous nonlinear widths. DeVore, Howard, and Micchelli prove worst-case lower bounds for approximation of smoothness balls when the encoder selecting N real parameters is continuous [14]. For Sobolev-type classes the characteristic norm-error rate is $N^{-s/d}$ under the theorem’s embeddings and parameter ranges. The result describes continuous encoders and a worst case over a whole function ball. Yarotsky maps an important complementary regime: deep ReLU networks whose weights may depend discontinuously on the target can improve the exponent in the weight count W by a factor of two for a fixed modulus of continuity [46]. Together, the results show that approximation rates depend on the parameter-selection model. A neural-training lower bound can build on them by establishing the continuity and function-class properties of the learned encoder/decoder.

Parametric redundancy. For regular identifiable k -dimensional parametric families, universal coding and Bayesian redundancy have a leading cumulative term $(k/2) \log n + O(1)$ under suitable compactness and Fisher-information assumptions [32, 33]. Per observation this is $O(k \log n/n)$. For singular and overparameterized neural models, an analogous result calls for a replacement of the regular-model dimension and Fisher-volume term; this is a natural point of contact with effective dimension and singular learning theory.

Statistical-query scope. Kearns introduced the statistical-query (SQ) model and showed that parity functions under the uniform distribution are not efficiently learnable by polynomially many inverse-polynomial-tolerance SQs [24]. This gives a clean algorithm-dependent lower bound. Applying it to neural training requires an explicit reduction of the optimizer to SQ access, usually through a noise, batching, or precision model, because finite-sample gradient descent can inspect individual examples. SQ and cryptographic lower bounds are distinct frameworks; stating the relevant oracle and computational assumption separately makes either connection precise.

6 Spectral and solvable models

6.1 Spectral inputs to kernel learning curves

For a kernel integral operator with eigenpairs (λ_k, ψ_k) and target $f^* = \sum_k c_k \psi_k$, the learning curve depends on the full pairs (λ_k, c_k^2) , as well as label noise and regularization. Bordelon et al. derive approximate typical-case modal learning curves, and Canatar et al. analyze spectral bias, task-model alignment, and possible non-monotonicity [6, 10]. Thus the eigenvalue decay $\lambda_k \asymp k^{-a}$ supplies one input, while the alignment profile c_k^2 , noise level, and regularization complete the exponent formula. This joint dependence is useful: targets with the same kernel but different alignment profiles provide direct experimental tests.

What can be said cleanly is a hard-cutoff heuristic, and it is worth stating carefully because the cutoff—not the tail sum—is where the algorithm enters. Suppose the learner resolves the modes up to some index k_* and leaves the rest untouched. The residual error is then the

unlearned target power, and with $c_k^2 \asymp k^{-b}$, $b > 1$,

$$\mathcal{E} \approx \sum_{k > k_*} c_k^2 \asymp k_*^{-(b-1)}. \quad (19)$$

For ridgeless kernel regression on n samples the cutoff is set by the effective ridge κ , which solves $n \asymp \sum_k \lambda_k / (\lambda_k + \kappa) \asymp \kappa^{-1/a}$; hence $\kappa \asymp n^{-a}$ and $\lambda_{k_*} \asymp \kappa$ gives $k_* \asymp n$. Samples buy modes roughly one for one, whatever the decay rate a , and Equation (19) gives $\mathcal{E}(n) \asymp n^{-(b-1)}$. Substituting the source/capacity parametrization $\lambda_k = k^{-\alpha}$, $c_k^2 = k^{-(1+2\alpha r)}$ recovers the familiar noiseless kernel rate $n^{-2\alpha r}$ [12].

The ridgeless and noiseless conditions specify this regime. Cui et al. show that label noise under optimal regularization gives the exponent $n^{-(b-1)/b}$ and a crossover from the noiseless rate as n grows [12]; the noiseless exponent also saturates once $b - 1 > 2a$. Spigler et al. measure spectrum-driven exponents on empirical data and find informative departures from teacher–student predictions for the same kernel [41]. The shared conclusions are spectral ordering—larger-eigenvalue modes are learned first—and dependence on alignment. Equation (19) becomes a learning curve once the algorithm-specific cutoff is supplied.

6.2 A rigorous linear-regression scaling law

Lin, Wu, Kakade, Bartlett, and Lee study infinite-dimensional linear regression trained by one-pass SGD, with a fixed Gaussian sketch of the covariates playing the role of model size and covariance eigenvalues decaying as $\lambda_k \asymp k^{-a}$, $a > 1$ [25]. In their notation the sketch dimension is M and the sample size is N ; translating to the conventions of Section 1, with N parameters and D samples, their source condition is $c_k^2 \asymp k^{-b}$ in the notation of Section 6.1 and the reducible test error is

$$\Theta\left(N^{-(b-1)} + D^{-(b-1)/a}\right), \quad 1 < b < a + 1, \quad (20)$$

which reduces to $\Theta(N^{-(a-1)} + D^{-(a-1)/a})$ under their isotropic parameter prior, the case $b = a$.

Equation (20) is two instances of the cutoff template (19), and reading it that way shows exactly where the algorithm enters. Sketching to N dimensions truncates at $k_* = N$, giving $N^{-(b-1)}$. One-pass SGD, unlike ridge regression, resolves a mode once the accumulated step size times its eigenvalue is order one, so $D\lambda_{k_*} \gtrsim 1$ and $k_* \asymp D^{1/a}$, giving $D^{-(b-1)/a}$. The same spectrum and the same target thus yield a data exponent of $(b-1)/a$ under one-pass SGD and $(b-1)$ under ridgeless kernel regression: the exponent is a property of the algorithm as much as of the data.

Two further points are worth extracting. The result resolves a puzzle—naively the variance error grows with model size, but here the implicit regularization of one-pass SGD makes that term strictly higher order, preserving monotone improvement in N . Both exponents are set by the pair (a, b) rather than being free: $\alpha_N = b - 1$ and $\alpha_D = (b - 1)/a$, hence $\alpha_N = a\alpha_D$. This is a falsifiable constraint linking two quantities that Chinchilla-style fits estimate independently. The Hoffmann exponents $\alpha_N = 0.34$ and $\alpha_D = 0.28$ correspond to $a \approx 1.2$ and $b \approx 1.3$, inside the admissible range. Measuring a transformer’s feature spectrum and alignment would turn this numerical consistency into a direct test. Equation (20) is therefore a useful joint scaling theorem with clearly specified linear-model, prior, and algorithmic assumptions.

6.3 Training dynamics and compute-optimal phases

Maloney, Roberts, and Sully provide a solvable random-feature model in which a power-law spectrum produces model-, data-, and compute-scaling regimes [26]. Bordelon, Atanasov, and Pehlevan add gradient-descent dynamics and explain why training-time and width exponents can differ, why early- and late-time width corrections can have different powers, and how repeated reuse of data builds a train–test gap [5]. Paquette et al. analyze a related one-pass-SGD

model across all iteration counts and derive four principal compute-optimal phases plus three subphases [30]. The phase boundaries depend on data complexity, target complexity, optimizer noise, and the random feature embedding. Together these solvable models show how optimization dynamics can choose both an exponent and a compute-optimal allocation from the same underlying spectral structure. Extending their state variables to learned, evolving representations is the next bridge to finite transformers.

7 Discrete tasks, memorization, and superposition

7.1 Associative memories with heavy-tailed inputs

Cabannes, Dohmatob, and Bietti analyze an outer-product associative-memory model with high-dimensional embeddings and a heavy-tailed distribution over discrete inputs [8]. They derive precise sample- and parameter-scaling laws for several estimators, including optimization-based ones, and validate the predicted regimes numerically. The model is valuable because it combines a Zipf-like source with an explicit finite-dimensional memory rather than assuming that the most useful items are simply stored. Its connection to transformer key-value memories is mechanistic, while its theorems are exact for the specified associative-memory model. It therefore supplies a rigorous intermediate step between Hutter’s pure memorizer and the quantization account below.

7.2 Quantization model

Michaud, Liu, Girit, and Tegmark propose that knowledge decomposes into discrete “quanta” learned roughly in decreasing order of frequency [27]. If quantum utilities obey $U_i \asymp i^{-q}$ with $q > 1$ and the first K quanta are mastered, then the remaining loss is the elementary tail

$$\sum_{i>K} U_i \asymp K^{-(q-1)}. \quad (21)$$

If, in addition, $K \asymp N$, this gives $L(N) - E \asymp N^{-(q-1)}$. The tail calculation is exact, and it is Equation (19) again with quantum utilities in place of target power and $k_* \asymp N$ as the supplied cutoff. The decomposition into quanta, learning order, independence, uniform capacity cost, and relation $K \asymp N$ are the model’s testable ingredients. A heavy-tailed utility ranking is plausible for language given Zipf-like frequency structure [47]; measuring skill frequencies separately from word frequencies would directly assess the transfer. The paper validates its predictions in toy settings and reports suggestive LLM evidence.

7.3 Hutter’s memorization model

Hutter analyzes a classifier on countably many discrete inputs that memorizes a label after first observing the input [19]. If input probabilities satisfy $\theta_i \propto i^{-(1+\eta)}$ with $\eta > 0$, the expected error over n samples scales as

$$\mathbb{E}E_n \asymp n^{-\eta/(1+\eta)}. \quad (22)$$

The mechanism is again a tail sum: inputs with $n\theta_i \gtrsim 1$ have almost surely been seen, and the error is the residual mass $\sum_{i>i_*} \theta_i$ with $i_* \asymp n^{1/(1+\eta)}$. Exponential input frequencies instead give an n^{-1} rate up to logarithmic factors, and finitely supported distributions eventually give exponential decay, cleanly isolating the role of the heavy tail. The result concerns this countable-input memorization rule; incorporating program length, execution time, or representation learning would define distinct extensions.

7.4 Feature superposition

Toy models of superposition show how sparse features can share representational dimensions [15]. For an illustrative random dictionary $W \in \mathbb{R}^{d \times M}$ with $M \gg d$ approximately isotropic unit columns and independent centred feature activations of variance $p(1-p)$ (as for a Bernoulli(p) feature that is active with probability p), linear readout of feature i has interference variance

$$\mathbb{E} \left| \sum_{j \neq i} \langle w_i, w_j \rangle x_j \right|^2 = p(1-p) \sum_{j \neq i} \langle w_i, w_j \rangle^2 \approx p(1-p) \frac{M}{d}. \quad (23)$$

The first equality uses independence and zero mean; the second uses $\mathbb{E} \langle w_i, w_j \rangle^2 \approx 1/d$ for random unit columns. Johnson–Lindenstrauss-type constructions show that M vectors can be made nearly orthogonal in dimension $d = O(\log M)$ [22], which helps keep interference tolerable for sparse features. The variance formula additionally uses isotropy and independent activations. It quantifies a capacity–interference tradeoff at rate M/d for a fixed random dictionary. Adding learned feature selection, a feature-importance distribution, and a map from representation error to next-token loss would turn this capacity mechanism into an end-to-end scaling theory; the importance tail would then connect naturally to Sections 7.2 and 7.3.

8 Comparative synthesis

Framework	Resource and assumptions	Principal result	Connection to neural scaling
Gaussian rate–distortion	Bit rate; Gaussian source; squared error; power-law covariance	$D(R) \asymp R^{-\rho}$ for $\lambda_k \asymp k^{-(1+\rho)}$ (Thm. 3.1)	Specify parameter precision, coding map, and distortion
Metric entropy/minimax	Samples; specified smoothness class, metric, and observation model	Rates such as $n^{-2s/(2s+d)}$ under regular conditions	Identify an effective language-model class and metric
Language horizon	Tokens; entropy and covariance decay; fast within-horizon learning	$\alpha_D = \gamma/(2\beta)$ in the horizon-limited regime	Extend the measured-statistics method to N and C
Manifold tiling	Cells proportional to parameters; smooth target and loss	Piecewise-affine mechanism gives $N^{-4/d}$	Measure effective cells and scale-dependent geometry
Kernel spectra	Samples; eigenpairs, target projections, noise, regularization	Ridgeless noiseless cutoff $k_* \asymp n$ gives $n^{-(b-1)}$	Track spectra and alignment during feature learning
Linear and dynamical models	Features, samples, steps; spectral source and target structure	Joint N, D law and multiple compute-optimal phases	Extend solvable dynamics to evolving representations
Discrete memory	Heavy-tailed input or utility; finite memory or learned quanta	Exact associative-memory rates and tail-sum mechanisms	Measure skill frequencies and learned memory allocation
Superposition	Width; sparse independent features; isotropic dictionary	Interference can scale as M/d in the toy model	Link learned features and interference to predictive loss

Read across the rows, one productive pattern emerges. In many analytic frameworks, the exponent traces back to the tail of a ranked quantity—covariance eigenvalues, kernel eigenvalues and target projections, cell counts on a manifold, quantum utilities, input frequencies, or token correlations—together with an algorithm-specific resolution rule. The controlled graph experiments of Barkeshli et al. show that this is not the only route to finite-range scaling. Measuring

the ranked quantity, intervening on data structure, and tracking representation dynamics alongside loss curves can distinguish the mechanisms.

9 Research opportunities

1. **Resource identification.** What operational quantity should replace parameter count in an information-theoretic model: quantized description length, effective rank, Fisher information, algorithmic complexity, or something else?
2. **Feature learning beyond kernels.** Extending fixed-kernel theory to predict how spectra and target alignment evolve during finite-width training would connect the strongest spectral results to representation learning.
3. **Dependence and data quality.** Tokens are neither independent nor equally informative. Pruning by example difficulty can improve power-law data scaling [40], while repeated data has a predictable diminishing-return curve [29]. These results motivate an operational “effective data” resource that includes quality, novelty, and dependence.
4. **Joint non-asymptotic theory.** Finite-range error bars across N, D, C , architecture, optimizer, and context would explain observed breaks, plateaus, and crossovers [7] and predict where regime changes occur.
5. **Loss versus downstream behavior.** The relation between smooth log-loss improvement and thresholded task success needs explicit measurement models [35], especially for rare skills and multi-step reasoning where a single token error can fail an entire trajectory.
6. **Test-time computation.** Search, sampling, verification, and external tools create an inference resource that can substitute for pretraining compute on some tasks [39]. A joint theory of pretraining and inference compute could extend $C \simeq 6ND$ into a full lifecycle allocation rule.

10 Conclusion

Power laws arise naturally when a problem contains scale-free structure: a polynomial spectral tail, a nonparametric smoothness class, long-range correlations, or a heavy-tailed frequency distribution. Solvable dynamical models and controlled graph experiments add optimization- and architecture-dependent routes. The central task is to identify which structure controls a particular trained neural network and how the learning algorithm exploits it.

Gaussian reverse water-filling, minimax estimation, kernel theory, linear SGD, and associative memory provide exact or controlled-model statements. Manifold geometry, quantization, and superposition provide complementary mechanisms with testable assumptions. The 2026 language-statistics theory connects a data exponent to corpus statistics measured independently of the learning curve and offers a clear template for progress: name the governing quantity, measure it, predict an exponent, and test that prediction on new corpora. Solvable dynamical theories provide a parallel template for parameter and compute scaling.

Two practical consequences follow. First, a fitted exponent is best reported together with its scale range, data mixture, optimization protocol, and uncertainty; the fitted floor E should likewise be distinguished from a corpus entropy rate. Second, comparing mechanisms requires measuring their primitives—spectra, alignments, intrinsic dimensions, correlation decay, frequency tails, and optimization regimes—alongside the loss curve. Neural scaling laws are robust empirical regularities supported by a growing collection of mathematical mechanisms. Their diversity is a strength: it supplies multiple, experimentally distinguishable paths toward a predictive theory.

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